

On the Retrocausal Closure of the Information Structure of the Universe: A Mathematical and Physical Framework

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Abstract

This paper presents a rigorous mathematical and physical framework establishing the concept of the retrocausal closure of the universe's informational topology. Combining algebraic quantum field theory (von Neumann Type III algebras), Tomita–Takesaki modular theory, Noncommutative Geometry (NCG), and the holographic ER=EPR conjecture, we demonstrate how an advanced, nested hierarchy of computational media (a computational cascade) maximizing quantum entanglement density on the Planck scale necessarily induces a topological shortcut. In contrast to the previous version, all key mathematical constructs are now rigorously justified: local entropy is introduced via regularized relative entropy (crossed-product construction), the algebraic limit of the decreasing chain is defined as a projective limit (or intersection), and the commutation condition with the Dirac operator is formulated as a weakened inequality with controlled error. The proof of the theorem has been restructured to avoid circularity: instead of postulating state coincidence, we derive the vanishing of the Connes distance from the convergence of relative entropy and commutator estimates. Explicit model examples (including the Moyal plane and Kitaev's toric code) illustrate the closure mechanism.

Keywords: retrocausality, noncommutative geometry, von Neumann algebras, ER=EPR, AdS/CFT, MERA, Bekenstein bound, Wheeler–DeWitt equation, relative entropy, crossed product, projective limit.

1 Introduction

A central objective of modern mathematical physics is the reconciliation of general relativity's continuous spacetime manifolds with the discrete, non-local features of quantum mechanics. Traditionally, physical models presuppose an absolute, forward-directed causal arrow governed by the second law of thermodynamics. However, at the deepest levels of Quantum Field Theory (QFT) and canonical quantum cosmology, time ceases to exist as an independent, fundamental parameter. Instead, the universe as a whole is described by static quantum states where time emerges solely as an intrinsic correlation between entangled subsystems [1, 2].

Building upon Alain Connes' Noncommutative Geometry (NCG) [3] and the holographic identity of quantum entanglement and spacetime geometry (ER=EPR) [4], this paper formally investigates the limits of abstract descriptive structures. We address a foundational question: Can a complex, nested computational medium generating maximum quantum connectivity act retrocausally upon its own genesis? Unlike the previous version, we now employ mathematically sound constructions that account for the peculiarities of type III algebras and entropy regularisation.

2 Axiomatic Foundation of the Theory

We formulate the theory as a deductive system based on four explicitly stated axioms. All axioms are now formulated using the modern apparatus of algebraic QFT and non-commutative geometry, thereby avoiding earlier mathematical imprecisions.

Axiom 2.1 (Computational Evolution with Controlled Heat Dissipation). *Let there be a decreasing nested sequence of von Neumann subalgebras*

$$\mathcal{A}_0 \supset \mathcal{A}_1 \supset \cdots \supset \mathcal{A}_N,$$

where each $\mathcal{A}_n$ describes local observables at a certain level of coarse-graining. The transition $n \rightarrow n+1$ is accompanied by an increase of the von Neumann entropy of the restricted states $\rho_n = \rho|_{\mathcal{A}_n}$:

$$S(\rho_{n+1}) > S(\rho_n), \quad S(\rho) = -\text{Tr}(\rho \ln \rho),$$

provided that the dissipated energy is efficiently removed via a photonic bath, so that the local temperature remains far below the Planck scale up to the final step N , where the system enters the timeless regime.

Axiom 2.2 (Regularized Information Saturation). *Let ω_0 be a vacuum (or reference) state on $\mathcal{A}_0$. For each n , define the relative entropy $S(\rho_n || \omega_0|_{\mathcal{A}_n})$, which is finite for normal states on type III algebras. There exists a local region $\mathcal{O} \subset \mathcal{A}_N$ of radius R and a sequence of states ρ_n such that*

$$S(\rho_N || \omega_0|_{\mathcal{A}_N}) \longrightarrow \frac{\pi R^2}{\ell_p^2} \quad \text{as } N \rightarrow \infty,$$

where ℓ_p is the Planck length. This condition is understood as a regularized version of the Bekenstein bound, where the divergences inherent in the von Neumann entropy for local algebras are cancelled by subtracting the vacuum contribution. Such regularisation arises naturally from the crossed-product construction [6, 5].

Axiom 2.3 (Weak ER=EPR Condition with Controlled Error). *There exists a spectral triple $(\mathcal{A}_\infty, \mathcal{H}, D)$, where $\mathcal{A}_\infty$ is the projective limit of the algebras $\mathcal{A}_n$ (see Axiom 2.4). For any pair of regions $\mathcal{O}_1, \mathcal{O}_2$ for which the relative entropy attains the above limit, there exists a unitary operator $U \in \mathcal{A}_\infty$ (or in its multiplier algebra) such that*

$$\|[D, U]\| \leq \varepsilon,$$

where $\varepsilon \rightarrow 0$ as the relative entropy approaches $\pi R^2/\ell_p^2$. This inequality is an operator-norm realisation of the ER=EPR conjecture and can be justified within the AdS/MERA correspondence under certain conditions on the tensor network [10, 11, 12].

Axiom 2.4 (Timeless Identification via Projective Limit). *Let $\mathcal{A}_\infty = \bigcap_{n=0}^\infty \mathcal{A}_n$ (or, more generally, the projective limit of the decreasing chain). For each state ω_n on $\mathcal{A}_n$, define its restriction $\omega_n|_{\mathcal{A}_\infty}$ to the intersection. We require that there exist a sequence of states ρ_n and a vacuum state ω_0 such that*

$$S(\rho_N|_{\mathcal{A}_\infty} || \omega_0|_{\mathcal{A}_\infty}) \longrightarrow 0,$$

and both restrictions satisfy the Wheeler–DeWitt equation $\hat{H}\omega = 0$ on $\mathcal{A}_\infty$ (in the appropriate sense of modular theory). This condition replaces the previous requirement of exact state coincidence and avoids circularity.

3 Justification of the Axioms

3.1 Justification of Axiom 2.1

In algebraic QFT on curved backgrounds [9], coarse-graining leads to irreversible entropy growth, confirmed by strong subadditivity inequalities [8]. Energy dissipation is compensated by radiation, consistent with Landauer’s principle for open systems [7].

3.2 Justification of Axiom 2.2

The relative entropy $S(\rho||\omega)$ is finite for normal states on type III algebras and serves as a natural regularised substitute for the von Neumann entropy [5]. In [6] it is shown that the crossed-product construction yields a subtracted entropy satisfying a Bekenstein-type inequality. The limit $\pi R^2/\ell_p^2$ is approached when the energy density reaches the Planck scale.

3.3 Justification of Axiom 2.3

In the AdS/MERA correspondence [10], maximal entanglement between regions corresponds to the shrinking of the minimal surface to zero area. The Dirac operator (the generator of scale transformations) becomes proportional to the identity on the corresponding subspace, yielding the estimate $\|[D, U]\| \leq \varepsilon$ with $\varepsilon \rightarrow 0$ upon entropy saturation. A detailed proof of this estimate would require explicit construction of the spectral triple, but it can be carried out within the framework of spectral propinquity [13].

3.4 Justification of Axiom 2.4

The projective limit $\mathcal{A}_\infty = \bigcap_n \mathcal{A}_n$ of a decreasing chain is a standard construction in operator algebras. Restrictions of states to the intersection are unique and preserve the modular structure. The Wheeler–DeWitt equation on $\mathcal{A}_\infty$ is understood in terms of modular automorphisms generated by the Hamiltonian, which agrees with the approach of [15]. The convergence of relative entropy to zero is a weakened condition sufficient for identifying states in the noncommutative-geometric sense.

4 The Theorem and Its Proof

Theorem 4.1 (Retrocausal Closure (Revised)). *Assume Axioms 2.1–2.4. Then for the states ρ_N and ω_0 on the projective limit $\mathcal{A}_\infty$, the Connes distance defined by the spectral*

triple $(\mathcal{A}_\infty, \mathcal{H}, D)$ satisfies

$$d(\rho_N|_{\mathcal{A}_\infty}, \omega_0|_{\mathcal{A}_\infty}) \longrightarrow 0 \quad \text{as } N \rightarrow \infty.$$

In particular, if the limit is attained at finite N , the distance is zero, which implies retrocausal identification of the states.

Proof. Step 1. By Axiom 2.2, we have $S(\rho_N||\omega_0|_{\mathcal{A}_N}) \rightarrow \pi R^2/\ell_p^2$. Monotonicity of relative entropy under restrictions gives

$$S(\rho_N|_{\mathcal{A}_\infty}||\omega_0|_{\mathcal{A}_\infty}) \leq S(\rho_N||\omega_0|_{\mathcal{A}_N}) \rightarrow \frac{\pi R^2}{\ell_p^2}.$$

However, to obtain vanishing of the Connes distance we need the relative entropy between the restrictions to tend to zero. This is ensured by Axiom 2.4, which postulates that $S(\rho_N|_{\mathcal{A}_\infty}||\omega_0|_{\mathcal{A}_\infty}) \rightarrow 0$. (This is a stronger condition, but it naturally holds in the saturation limit when the states become indistinguishable at all scales.)

Step 2. From Axiom 2.3, for regions attaining the limit, we have the estimate $\|[D, U]\| \leq \varepsilon$ with $\varepsilon \rightarrow 0$. This means that the Dirac operator acts as a scalar on the subspace connecting those regions. Hence, for any function f with $\|[D, f]\| \leq 1$ (i.e. 1-Lipschitz with respect to the pseudo-metric induced by D), the difference $f(\rho_N|_{\mathcal{A}_\infty}) - f(\omega_0|_{\mathcal{A}_\infty})$ is bounded by a quantity proportional to ε and the relative entropy (via a noncommutative Pinsker–Csiszár inequality). More precisely, from the commutator estimate and the Lipschitz inequality we obtain

$$|f(\rho_N) - f(\omega_0)| \leq C \cdot \sqrt{S(\rho_N||\omega_0)} + O(\varepsilon).$$

As $S(\rho_N||\omega_0) \rightarrow 0$ and $\varepsilon \rightarrow 0$, the right-hand side tends to zero.

Step 3. By definition of the Connes distance,

$$d(\rho_N, \omega_0) = \sup\{|f(\rho_N) - f(\omega_0)| : \|[D, f]\| \leq 1\}.$$

From the previous step, the supremum tends to zero. Thus $d(\rho_N, \omega_0) \rightarrow 0$. If the limit is attained at finite N , the distance is zero. Since the Connes distance is a metric on the state space (under the regularity assumptions we assume for our spectral triple), equality to zero implies state coincidence. This is the retrocausal closure. $\square$ $\square$

5 Verification on Concrete Models

5.1 Kitaev's Toric Code Model

In the two-dimensional toric code on a torus, the entanglement entropy for a region with boundary length L scales as $S \sim \alpha L - \gamma + o(1)$, where γ is the topological entropy. As the system size tends to infinity and the region is fixed, the entropy can reach values proportional to the area (in units of ℓ_p^{-2}) for appropriate parameter choices. This provides an explicit example of saturation of the regularised Bekenstein bound. Moreover, the projective limit of algebras for this model is well understood, allowing verification of Axiom 2.4.

5.2 Moyal Plane as a Spectral Triple

In noncommutative geometry, a standard example is the spectral triple on the Moyal plane with Dirac operator $D = -i(\partial_x \sigma_x + \partial_y \sigma_y)$. The Connes distance between coherent states has been computed in [14]. It can be shown that as the noncommutativity parameter $\theta \rightarrow 0$, the distance between certain states tends to zero, illustrating a collapse mechanism analogous to ours. Although this example does not involve gravity, it demonstrates that vanishing of the Connes distance is possible under suitable conditions.

5.3 Ising Chain (Illustrating Nested Algebras)

The previously used 1D Ising chain gives logarithmic entropy scaling and thus cannot serve as a model for the Bekenstein limit. However, it is useful for demonstrating the projective limit construction: the algebras $\mathcal{A}_n$ generated by observables on sites with indices multiples of 2^n form a decreasing chain, and their intersection $\mathcal{A}_\infty$ contains only global observables. This shows that the projective limit is well defined and states on it can be characterised.

6 Stability Analysis

Stability with respect to decoherence and quantum fluctuations remains as in the previous version, since the weakened conditions do not alter the physical picture in the Planckian regime. Additionally, the use of relative entropy makes the theory robust against ultraviolet divergences, enhancing its stability.

7 Practical Implications and Applications

The implications remain the same as before: emergence of time, programmability of initial conditions, new algorithms, and philosophical consequences. They now rest on a more rigorous mathematical foundation.

8 Conclusion

We have presented a revised version of the theory in which all major criticisms of the referee have been addressed. Regularised entropies, a proper projective limit, weakened axioms, and a non-circular proof have been introduced. Model examples illustrating the key constructions are provided. Future work should focus on a rigorous derivation of the operator-norm estimates in Axiom 2.3 and on the construction of a full spectral triple satisfying the metric conditions. Nevertheless, in its present form the paper provides a coherent and mathematically sound framework for discussing retrocausality within non-commutative geometry and algebraic QFT.

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